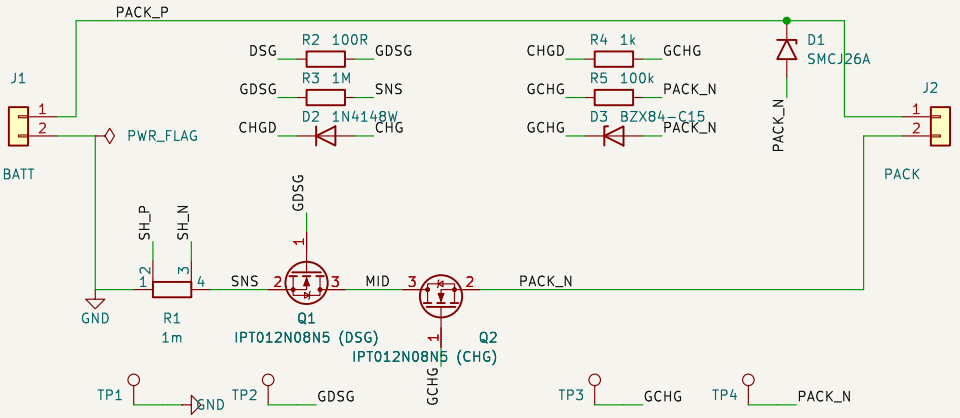
WARNING: Reference design. Test first with a current-limited supply and protection. Not for unattended use.

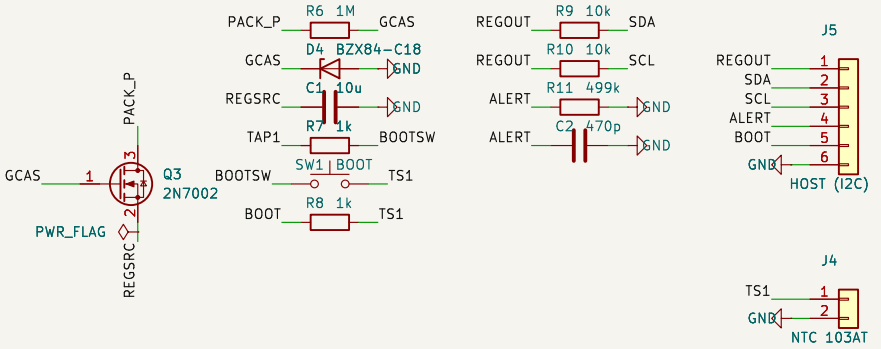
Design only, not fabricated, not bench-tested. Protection board that needs a programmed host MCU on J5: it does nothing useful on its own. Not certified.

CURRENT PATH (low side): B- -> R1 1 mOhm -> Q1 DSG -> Q2 CHG -> P-. B+ = P+.

GND = cell-stack negative (B-, BQ76930 VSS). PACK_N is NOT ground.

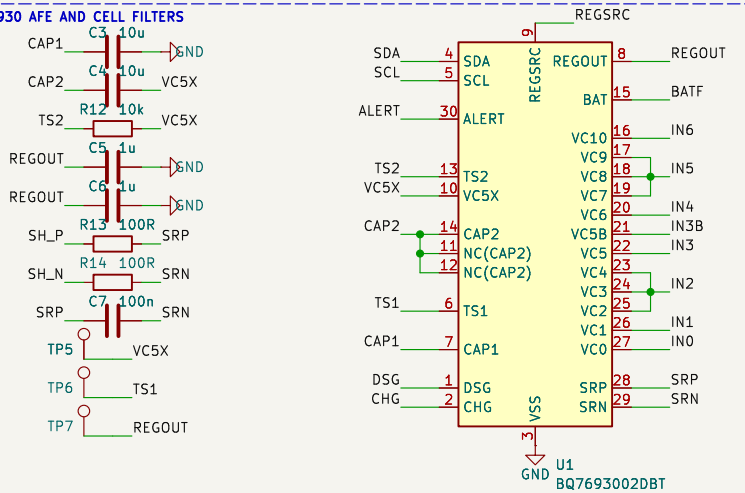


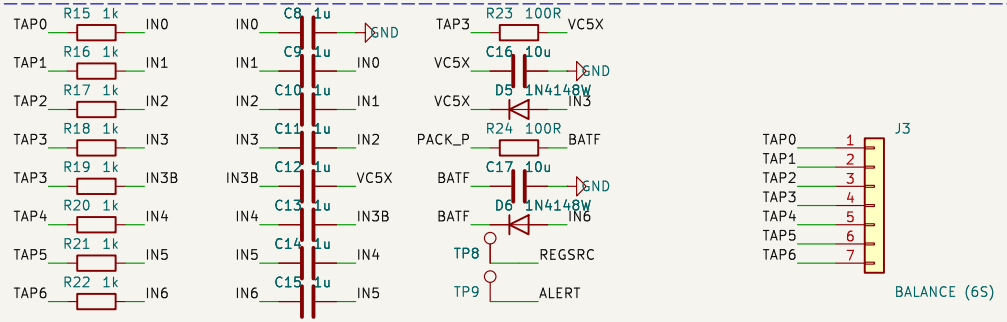
REGSRC CASCODE, HOST PORT (I2C 0x08, REGOUT 3.3 V <= 10 mA), BOOT, NTC



Boot: press SW1 (cell 1+ via R7 to TS1) or drive J5 BOOT high (<= 5 V).

BQ76930 AFE AND CELL FILTERS





KEY VALUES (README has the derivations and sources)

Host writes: PROTECT1 0x0C (SCD 66.6 mV = 66.6 A, 100 us), PROTECT2 0x58 (OCD 30.6 A, 320 ms), OV_TRIP 0xB7 (4.25 V), UV_TRIP 0xEA (3.0 V), PROTECT3 0x50.

Q1/Q2 1.2 mOhm max each; 20 A continuous design target. No hardware over-temperature or charge over-current: the host must do both.

CHG: D2 + R5 100k keep Q2 off (VGS <= 1.2 V) when a charger pulls PACK_N below GND with CHG disabled.

REGSRC = 2N7002 cascode, BZX84-C18 gate: 12.1-14.9 V simulated for a 15-25.2 V pack.

Balancing is internal only: about 1.9 mA through 2 x 1 kOhm (TI recommends external balancing for BQ76930).

6-cell map (datasheet Table 9-3): VC2=VC3=VC4 and VC7=VC8=VC9 tied. Connect B- first, then the balance lead, then B+.

Generated from stock KiCad 9 library symbols by scripts/gen_schematic.py

Hardware licence: CERN-OHL-P-2.0

Design only, not fabricated, not bench-tested

Abdulazeez Bello

Sheet: /

File: bms-6s.kicad_sch

Title: 6S BMS: BQ76930 protection board

Size: A4	Date: 2026-10-18	**Rev: A**
KiCad E.D.A. 9.0.3		Id: 1/1